

- The **Mid Term Exam 1- Part 1** for **STA 2023-0002** will be held in class from **10:30 AM to 11:45 AM** on **February 26, 2026**.
- The exam will cover Chapters 1-3.
- The exam is **closed-book and closed-notes**.
- You may use a **calculator** and **two double-page cheat sheet**.
- You are **not allowed** to use laptops, cellphones, tablets, or any other electronic devices.
- All cellphones must be **turned off** before the exam begins.
- Show all necessary steps for full credit.
- **Write all work clearly.** Answers without supporting steps may not receive credit and Round all probabilities to **four decimal places**.
- **Write answers in the spaces provided.** If you need additional space, clearly label the continuation.
- **Total points: 50.**

By signing below, I acknowledge that I have read and understood the instructions above and agree to comply with the university's policies on Academic Integrity for this exam.

Academic Integrity Statement: "I acknowledge that I have fully complied or will comply with all aspects of the university's policies on Academic Integrity and the exam instructions in submitting this exam."

Name: _____

Student ID: _____

Question 1 (5 points)

A nutritionist studies whether a new diet reduces cholesterol. She selects 80 participants and measures their cholesterol before and after the diet. After 12 weeks, the average drop is 15 points.

- (a) (1 point) Identify the **population**.
- (b) (1 point) Identify the **sample**.
- (c) (1 point) Identify the **individuals**.
- (d) (1 point) Identify the **parameter**.
- (e) (1 point) Identify the **statistic**.

Question 2 (12 points)

For each variable, state: 1) Qualitative or Quantitative 2) If Quantitative, Discrete or Continuous 3) Level of Measurement

- (a) **(3 points)** Type of car transmission (manual, automatic, CVT)

- Qualitative
- Not applicable (not quantitative)
- Nominal

- (b) **(3 points)** Number of pets owned

- Quantitative
- Discrete
- Ratio

- (c) **(3 points)** Temperature in Fahrenheit on a given day
- (d) **(3 points)** Ranking of the top 5 tennis players

Question 3 (4 points)

Researchers collect survey responses from college students about daily screen time and GPA.

- (a) **(1 point)** Is this study Observational or an Experiment?
Observational study
- (b) **(1 point)** Explanatory variable = _____
- (c) **(1 point)** Response variable = _____
- (d) **(1 point)** Can causation be claimed? Yes / No

Question 4 (3 points)

A company wants to know employee opinions on remote work. Consider the following scenarios and identify the type of bias (Sampling / Nonresponse / Response).

- (1) **(1 point)** Survey is sent only to employees at headquarters.
Sampling Bias
- (2) **(1 point)** Only 20% of employees respond, mostly those unhappy with remote work.
- (3) **(1 point)** Some employees give “safe” answers they think management wants.

Question 5 (8 points)

A survey asked 200 households their favorite weekend activity.

(a) (4 points) Fill in the missing relative frequencies and degree measures.

Activity	Frequency	Relative Frequency	Degree Measure
Sports	60	—	—°
Movies	50	—	90°
Shopping	40	0.20	—°
Reading	30	0.15	54°
Other	20	0.10	36°

(b) (1 point) Which activity is most popular?

Sports

(c) (1 point) What percent of households prefer Reading?

15%

(d) (2 points) True/False: More than half of households prefer Sports or Movies.

Question 6 (5 points)

Distribution of quiz scores for 40 students:

Interval	Frequency
40–49	4
50–59	6
60–69	10
70–79	12
80–89	8

(a) (1 point) What are the lower and upper class limits of the 3rd class?

Lower class limit = 60, Upper class limit = 69

(b) (1 point) What is the class width?

- (c) **(1 point)** What percent scored 70 or higher?
- (d) **(1 point)** How many scored below 60?
- (e) **(1 point)** Which class has the highest frequency?

Question 7 (4 points)

Identify the likely shape of each distribution (Bell-shaped, Uniform, Skewed Left, Skewed Right).

- (a) **(1 point)** Salaries of employees, where a few executives earn millions.
Skewed Right
- (b) **(1 point)** Heights of 10-year-old boys.
- (c) **(1 point)** Outcomes of rolling a fair die 200 times.
- (d) **(1 point)** Waiting times at a restaurant, where most are short but a few very long.

Numerically Summarizing Data: Use the Daily Commute Times data for questions 8-11

Daily Commute Times:

The following data represent the one-way commute times (in minutes) of 10 employees at a company:

18, 22, 25, 28, 30, 31, 33, 35, 36, 40

Hint: Make sure you know how to use your calculator to compute the mean and the population variance (standard deviation) for the below questions.

Question 8 (8 points)

- (a) (2 points) Find the population mean.

$$\sum x = 18 + 22 + 25 + 28 + 30 + 31 + 33 + 35 + 36 + 40 = 298$$

$$\mu = \frac{298}{10} = 29.8$$

Population mean = 29.8 minutes

- (b) (3 points) Find the population variance and population standard deviation.
- (c) (1 point) Find the population median.
- (d) (1 point) Find the population mode.
- (e) (1 point) Based on mean vs. median, describe the shape.

Question 9 (4 points)

Suppose we select the first 5 observations as a sample:

Sample: 18, 22, 25, 28, 30

Sample size: $n = 5$

- (a) (2 points) Find the sample mean.

- (b) **(2 points)** Find the sample variance.

$$\sum (x - 24.6)^2 = 95.2$$

$$s^2 = \frac{95.2}{4} = 23.8$$

Sample variance = 23.8

Question 10 (6 points)

- (a) **(3 points)** Find the five-number summary (minimum, Q1, median, Q3, maximum).
- (b) **(1 point)** Find the interquartile range (IQR).
- (c) **(2 points)** Find the lower fence and upper fence. Identify any outliers.

Question 11 (6 points)

Empirical Rule (68–95–99.7 Rule)

Use the population mean and population standard deviation calculated in Question 8.

- (a) **(2 points)** About 95% of commute times should lie between what two values?
- (b) **(2 points)** What percentage of commute times should fall between 23.4 minutes and 36.2 minutes?
- (c) **(2 points)** What percentage of commute times should fall between 36.2 minutes and 42.6 minutes?